

3.5 Вопрос Докажите формулу Вальмистеда.

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{2n} \right) \left(\frac{(2n)!!}{(2n-1)!!} \right) = \lim_{n \rightarrow +\infty} \left(\frac{2 \cdot 4 \cdot 6 \dots \cdot 2n}{1 \cdot 3 \cdot 5 \dots (2n-1)} \right) \cdot \frac{1}{2n}$$

$$4 \int_R \int_0^{\frac{\pi}{2}} \sin^n x dx = \left| \begin{array}{l} u = \sin^{n-1} x \\ du = (n-1) \sin^{n-2} x \cdot \cos x dx \end{array} \right. \quad \left. \begin{array}{l} dv = \sin x dx \\ v = -\cos x \end{array} \right|$$

$$\equiv \sin^{n-1} x \cos x \Big|_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x dx \equiv$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx =$$

$$= (n-1) \int_{n-2} - (n-1) \int_n \Rightarrow \int_n = \frac{n-1}{n} \int_{n-2}$$

$$\sin^{2n+1} x < \sin^{2n} x < \sin^{2n-1} x$$

$$x \in (0) \frac{\pi}{2}$$

$$y_{2n+1} < y_{2n} < y_{2n-1} \quad (*)$$

$$y_0 = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}$$

$$y_1 = \int_0^{\frac{\pi}{2}} \sin x \, dx = -\cos x \Big|_0^{\frac{\pi}{2}} = 1$$

$$y_2 = \frac{2-1}{2} y_0 = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

$$y_3 = \frac{3-1}{3} y_1 = \frac{2}{3}$$

$$y_4 = \frac{4-1}{4} y_2 = \frac{3}{16} \pi$$

$$y_6 = \frac{6-1}{6} \quad y_4 = \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \cdot \frac{\sqrt{2}}{3} = \frac{5!!}{6!!} \cdot \frac{\pi}{2}$$

$$y_{2n+1} < y_{2n} < y_{2n-1} \quad (*)$$

$$\frac{(2n)!!}{(2n+1)!!} < \frac{(2n-1)!!}{(2n)!!} \cdot \frac{\sqrt{2}}{2} < \frac{(2n-2)!!}{(2n-1)!!} \bigg/ \frac{(2n)!!}{(2n-1)!!}$$

$$\frac{1}{2n+1} \cdot \frac{(2n)!!}{(2n-1)!!} < \frac{\sqrt{2}}{2} < \frac{(2n)!!}{(2n-1)!!} \cdot \frac{1}{2n}$$

